رواد مصر الرقمية

وزارة الاتصـــــالات
وتكنولوجيا المعلومات

Adaptive Learning Assistant:

An AI-Powered Learning Chatbot for Dynamic Student Engagement

Hosam Abdelhady - Omnia Ghalwash - Mohamed Hussien - Doaa Abdelraouf



Personalized. Conversational. Learning alongside the student.

Presentation Agenda

What we'll cover today

01 | Problem Statement.

02 | Project Idea.

03 | Core Features.
What the Adaptive Learning Assistant do for students.

04 | System Architecture.

05 | Data Structure.

06 | Impact & Outcomes.

07 | Project`s Roadmap/ Roles.

08 | Deliverables/ Reports.

The Challenge:

To Adapt with The Needs of Modern Education.

- Student
- Teacher

Static courses

Fixed courses with traditional classrooms are not enough now to satisfy the student`s learning pace and style.

Resource Limitations

Students need 24-7 on-demand support. Tutors and teachers are often overloaded with tasks.

Lack of Continuous Engagement

Passive learning methods lead to less- engagement and low retention rates.

Lack of Stored Progress.

Students can not keep stored record of his daily progress or, personalized feedback.

60% of students feel understimulated

1:30 average teacher-to-student ratio

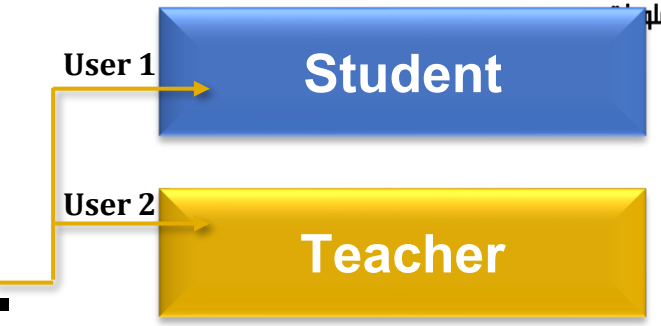
70% of learning gap due to no feedback

We need to save time and effort in the educational process for both main users (Student & Teacher)

OUR SOLUTION:

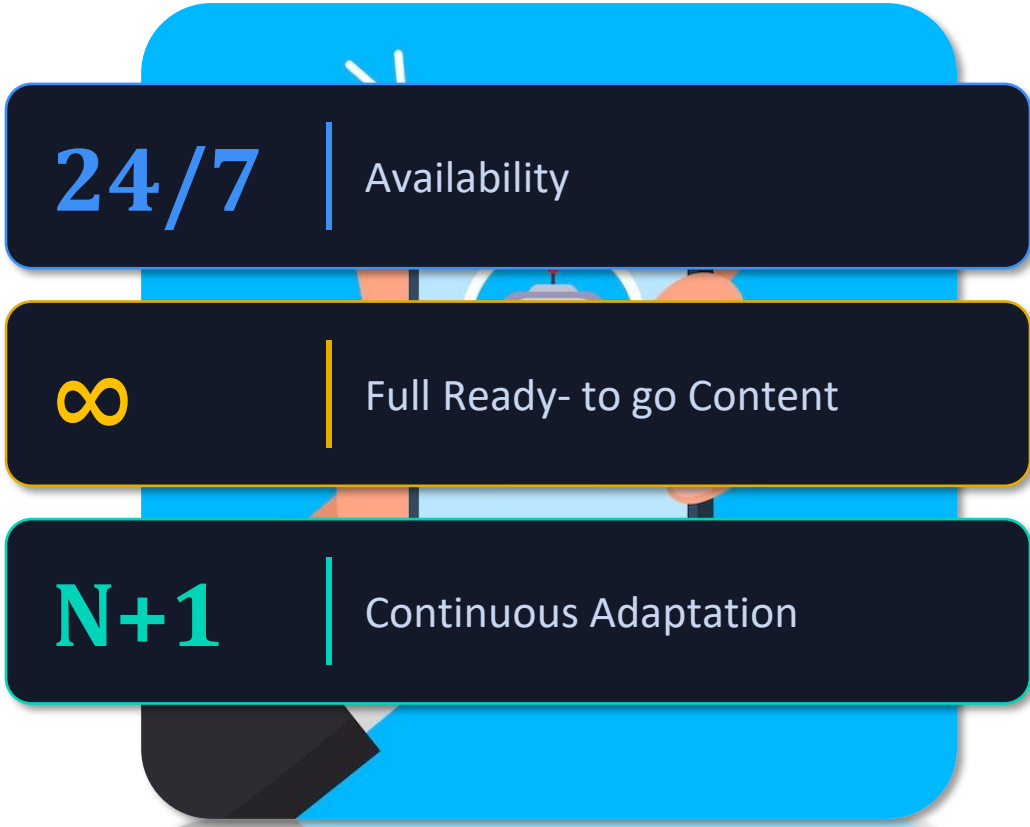
Meet the Adaptive Learning Assistant.

A conversational AI tutor that understands each student and reshapes his learning path in real time.



These all will help the teacher to reduce both effort and time consumed.

1. **Chats naturally: Ask The Tutor**
   Answers questions in plain language, 24/7, like a patient personal tutor.
2. **Learn with Us**
   Introduces an adaptive learning process through visual content, charts, and an easy friendly illustration.
3. **Get Ranked**
   Through an Interactive, adaptive quizzes with answer model and an explanation.
4. **Get a Feedback**
   Through a grading system to your uploaded assignments, Quizzes and daily progress.
5. **Keeps students engaged**
   Nudges, streaks, and goals turn studying into an active dialogue.



What can The Adaptive Learning Assistant do for students.

AI-Powered Dialogue

- Natural language Q&A.
- Context-aware conversations.
- Multi-resources expertise.

Adaptive Learning Paths

- Dynamic curriculum adjustment.
- A friendly Learning style.

Multi-Modal Content

- Text, image & video support
- Interactive quizzes
- Summarized study notes

Progress Testing

- Real-time performance tests.
- Strength & gap identification.
- Detailed model answers with explanation.

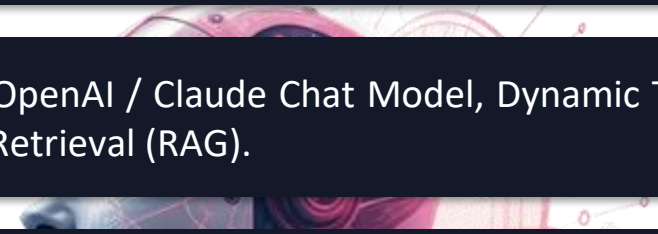
Grading Process/ Weakness points

- Immediate answer evaluation.
- Step-by-step explanations.
- Providing the student`s weakness points.



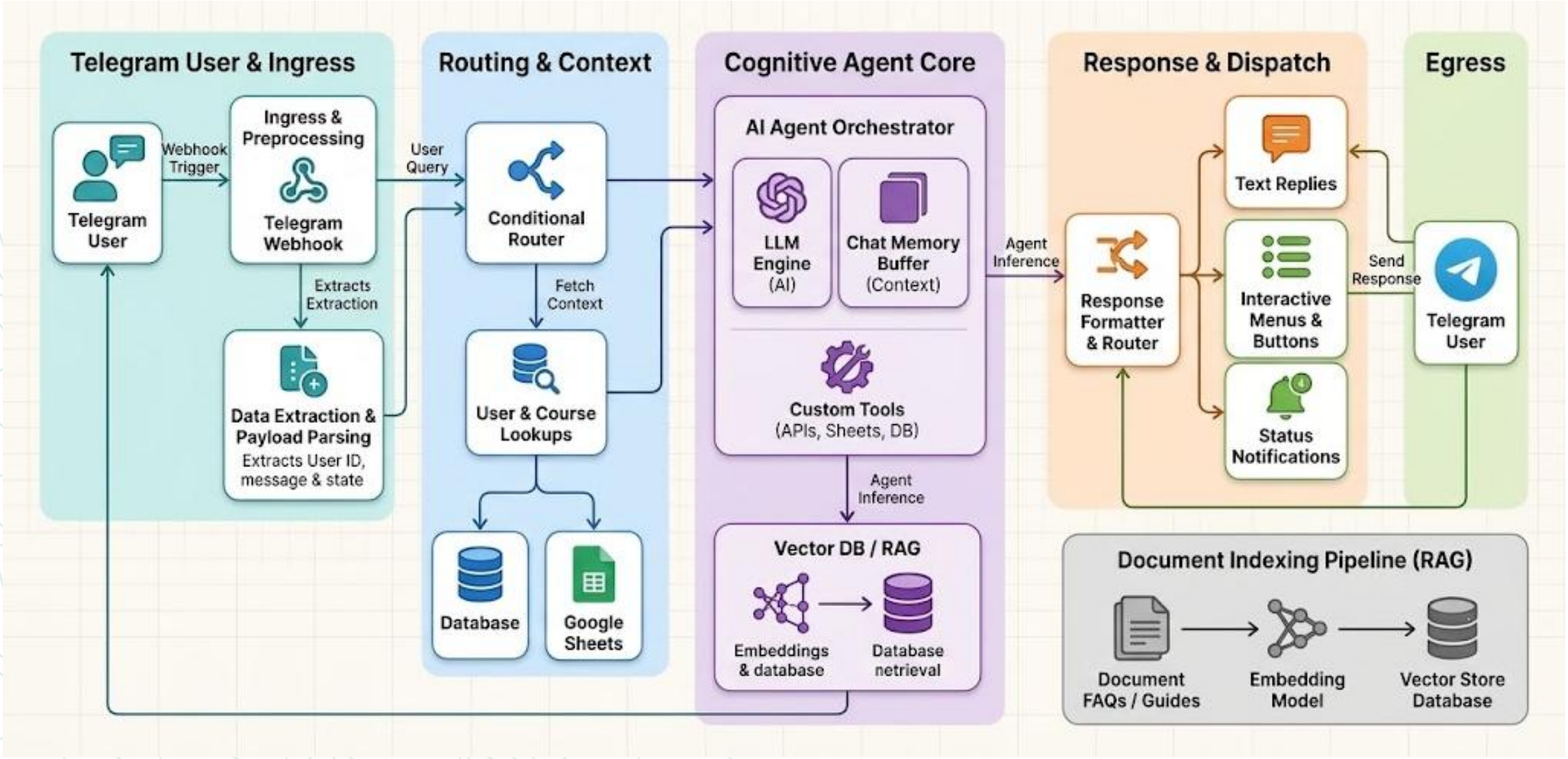
The System Architecture

Frontend Layer	Telegram Chatbot (Mobile / Desktop / Web).
Execution& API Gateway	n8n Workflow Engine (Webhook Trigger, Switch, IF, Code, Set Nodes).
Cognitive & AI Agent Layer	n8n AI Agent (LangChain Orchestrator), OpenAI / Claude Chat Model, Dynamic Tool Calling (SQL/Sheets Tools), Vector Store Retrieval (RAG).
Adaptive Core	Learning Model. Multi-Branch n8n Logic. LLM Grader. Postgres Adaptive State Engine.
Data & Storage	PostgreSQL / Supabase (pgvector Extension), LangChain Postgres Chat Memory, Google Drive Knowledge Base.





04 SYSTEM ARCHITECTURE



Data flow

off-pipeline

Knowledge Base RAG Ingestion Pipeline

Core Objective: a background pipeline responsible for storing and organizing the course material.

- **Trigger (Google Drive)/ File Loader Node:** Detects newly uploaded or modified curriculum documents (PDFs, lecture notes, slides) in the designated Google Drive folder.
- **Default Data Loader/ Splitter:** Extracts raw text and structural metadata, then Text Splitter Splits extracted text into overlapping semantic chunks to preserve context across boundaries.
- **Embeddings:** Google Gemini generates dense 768-dimensional semantic vectors for every document chunk.
- **Vector Database:** Supabase Vector Store is used to coordinate the ingestion of document chunks and metadata.

on-pipeline

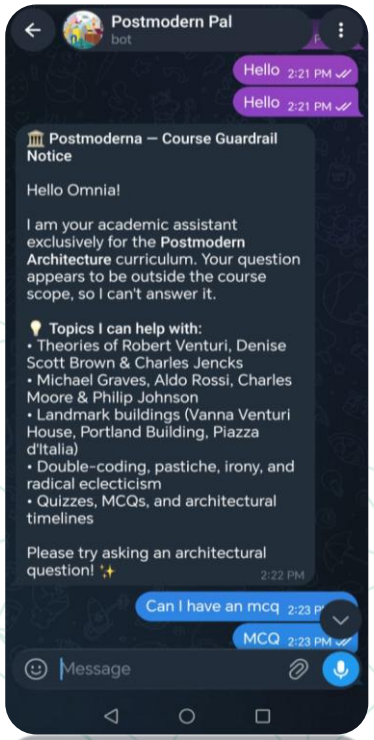
Real-Time Student Engagement & Assessment Engine

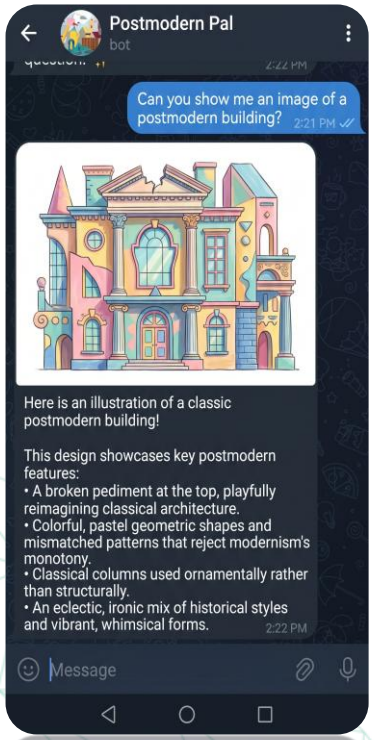
Core Objective: An event-driven operational engine that handles real-time student interaction, conversational AI tutoring, and automated assessment workflows.

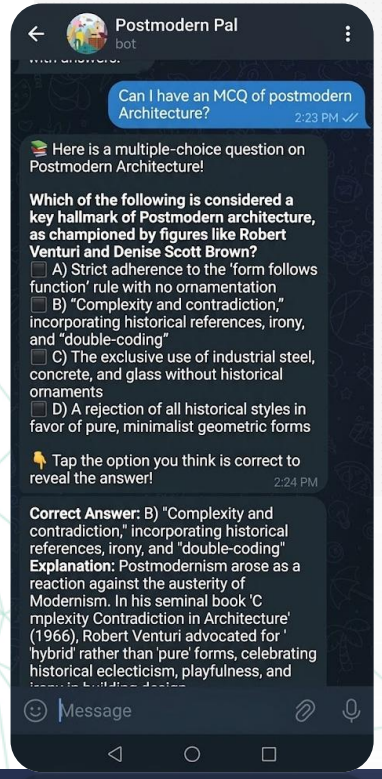
- **Entry & Pre-Processing:** The Telegram Trigger listens for real-time webhook POST payloads; extraction nodes normalize session variables; PostgreSQL queries student registration states; and a Master Switch node splits execution into three operational branches.
- **Branch 1 (Onboarding & State Management):** Handles first-time users (/start), profile updates, and state resets by updating student records in PostgreSQL and dispatching formatted greeting or dashboard messages via Telegram.
- **Branch 2 (Adaptive AI Tutor & RAG Agent Pipeline):** Houses the LangChain AI Agent powered by Google Gemini, utilizing Postgres Chat Memory for multi-turn conversation continuity and the Supabase Vector Store tool for semantic curriculum retrieval.
- **Branch 3 (Interactive Quiz & File Evaluation Pipeline):** Manages structured learning activities, interactive quizzes, and assignment file uploads; invokes an LLM grader to evaluate student responses against rubrics; updates scores in PostgreSQL; and routes execution conditionally between pass unlocks and remediation guidance.

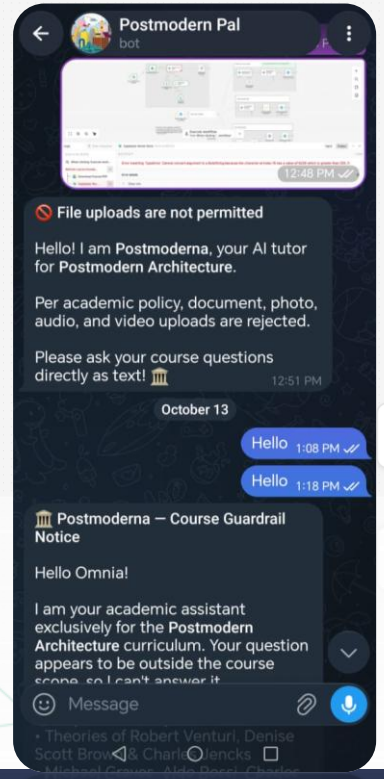
رواد مصر الرقمية

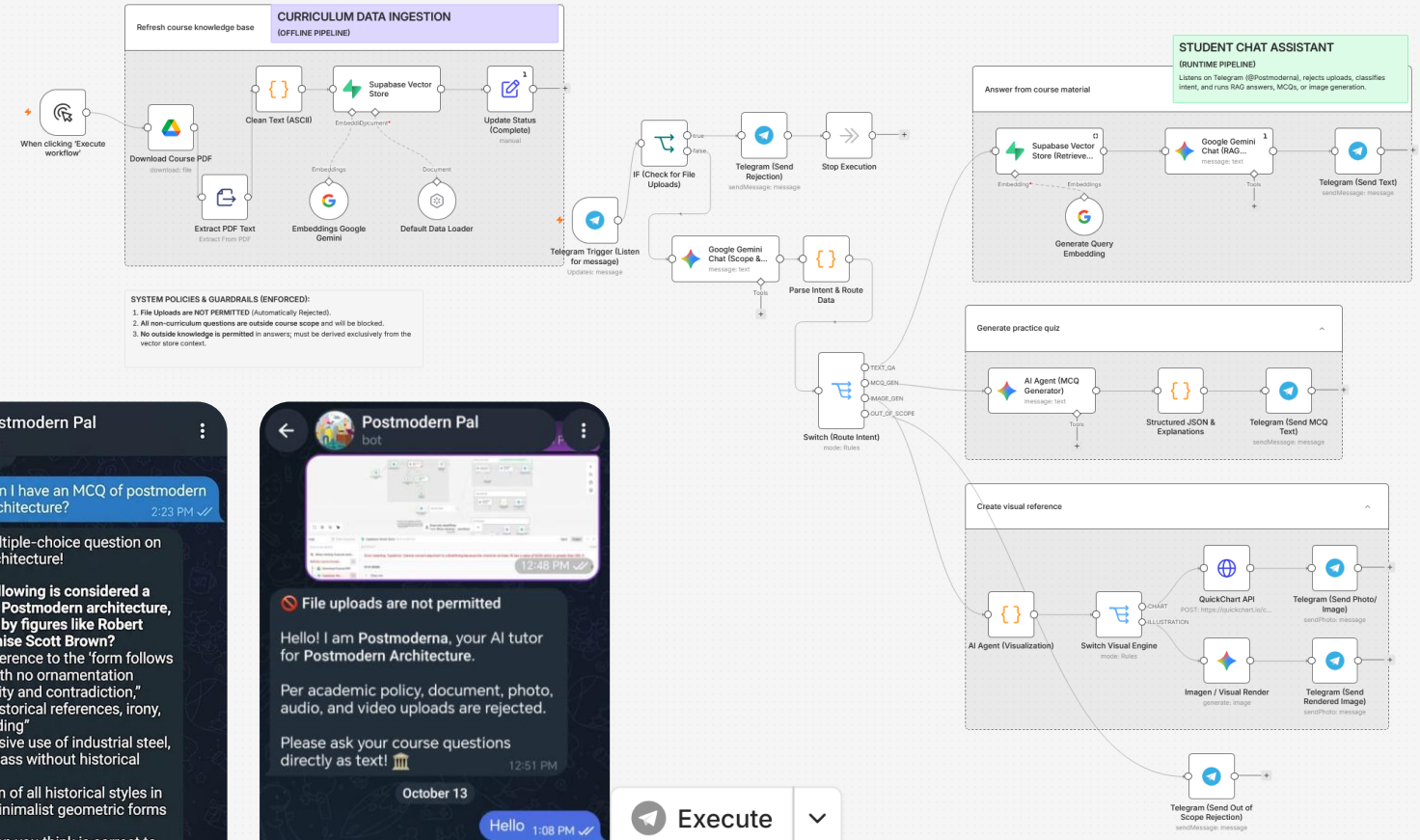
05 DATA STRUCTURE

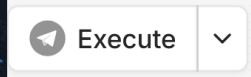




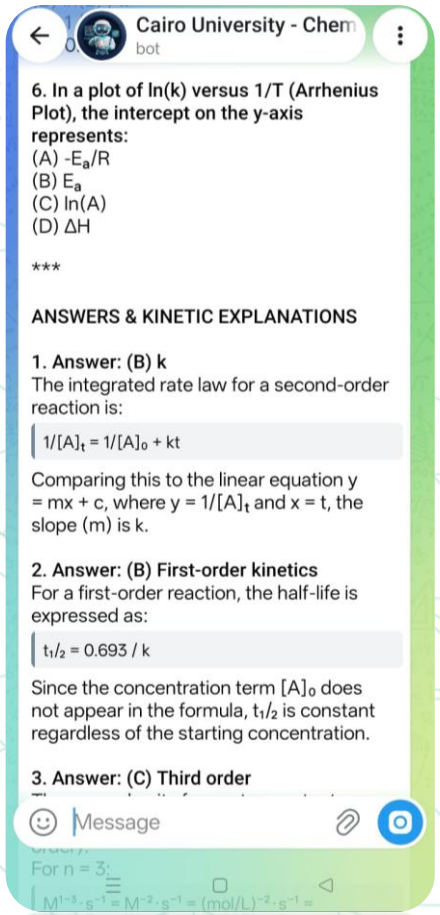


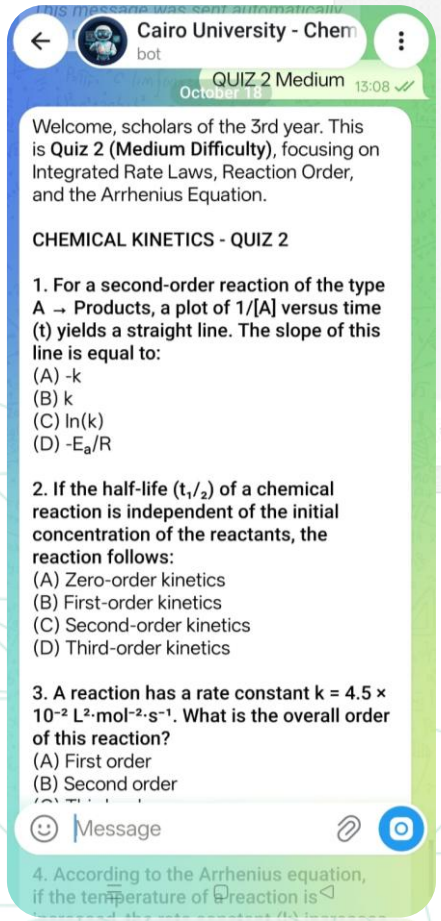


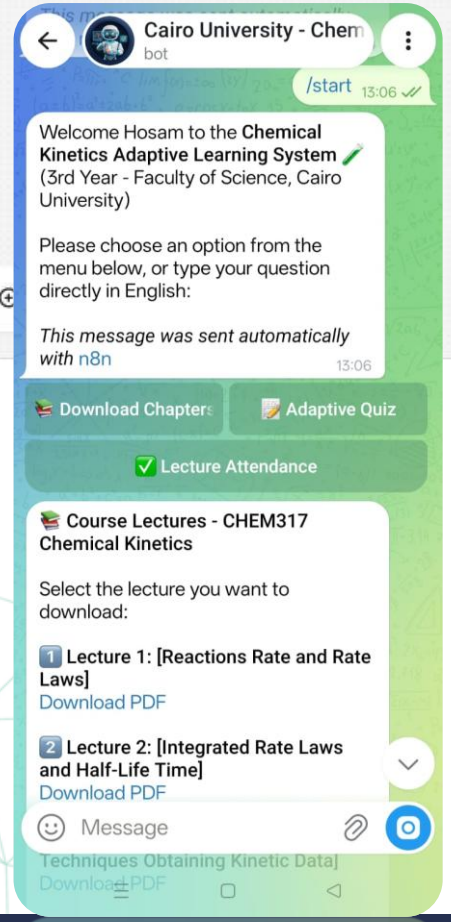


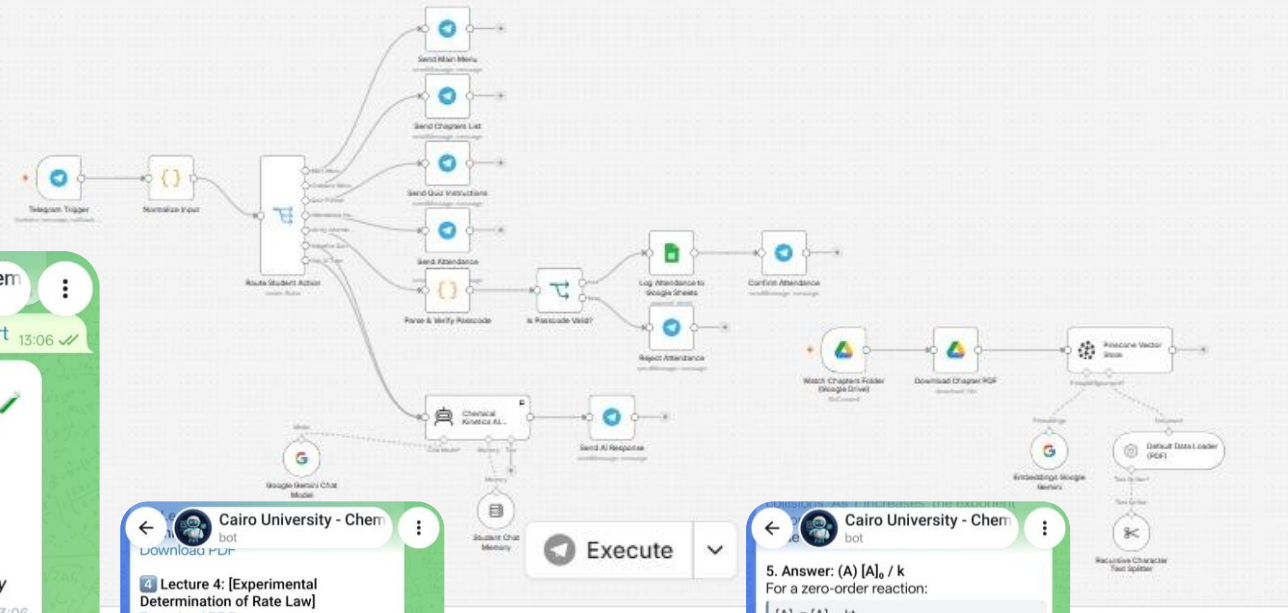


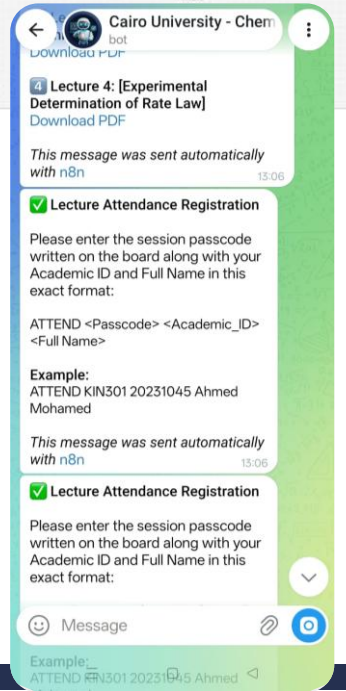
HOSSAM

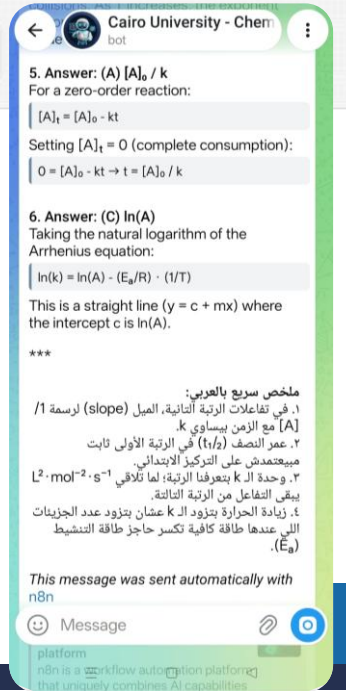

Cairo University - Chem
bot
6. In a plot of ln(k) versus 1/T (Arrhenius Plot), the intercept on the y-axis represents:
(A) -Ea/R
(B) Ea
(C) ln(A)
(D) ΔH
***
ANSWERS & KINETIC EXPLANATIONS
1. Answer: (B) k
The integrated rate law for a second-order reaction is:
1/[A]t = 1/[A]0 + kt
Comparing this to the linear equation y = mx + c, where y = 1/[A]t and x = t, the slope (m) is k.
2. Answer: (B) First-order kinetics
For a first-order reaction, the half-life is expressed as:
t1/2 = 0.693 / k
Since the concentration term [A]0 does not appear in the formula, t1/2 is constant regardless of the starting concentration.
3. Answer: (C) Third order
Message


Cairo University - Chem
bot
QUIZ 2 Medium 13:08
Welcome, scholars of the 3rd year. This is Quiz 2 (Medium Difficulty), focusing on Integrated Rate Laws, Reaction Order, and the Arrhenius Equation.
CHEMICAL KINETICS - QUIZ 2
1. For a second-order reaction of the type A → Products, a plot of 1/[A] versus time (t) yields a straight line. The slope of this line is equal to:
(A) -k
(B) k
(C) ln(k)
(D) -Ea/R
2. If the half-life (t1/2) of a chemical reaction is independent of the initial concentration of the reactants, the reaction follows:
(A) Zero-order kinetics
(B) First-order kinetics
(C) Second-order kinetics
(D) Third-order kinetics
3. A reaction has a rate constant k = 4.5 × 10⁻² L²·mol⁻²·s⁻¹. What is the overall order of this reaction?
(A) First order
(B) Second order
Message

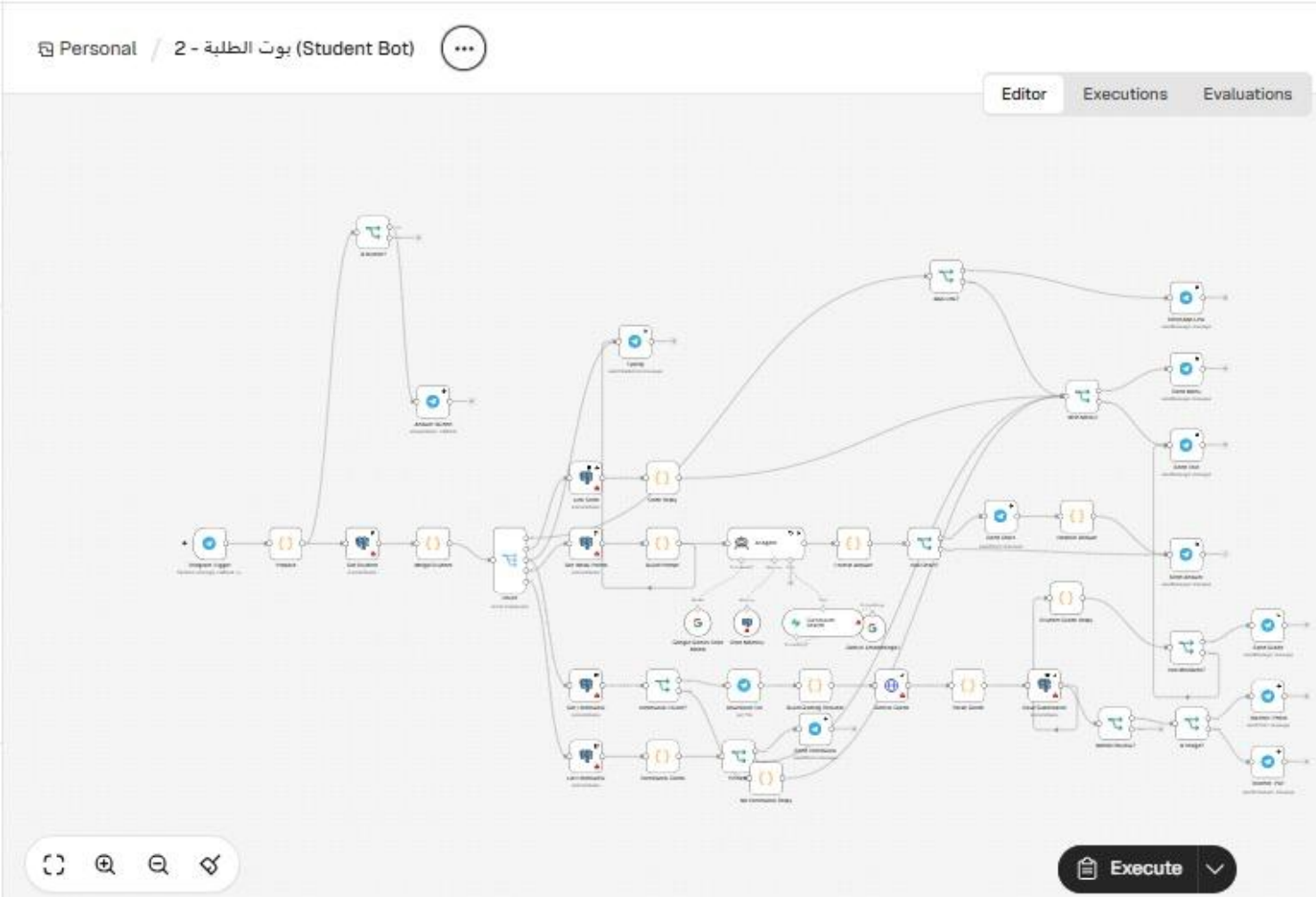

Cairo University - Chem
bot
/start 13:06
Welcome Hosam to the Chemical Kinetics Adaptive Learning System
(3rd Year - Faculty of Science, Cairo University)
Please choose an option from the menu below, or type your question directly in English:
This message was sent automatically with n8n 13:06
Download Chapters
Adaptive Quiz
Lecture Attendance
Course Lectures - CHEM317 Chemical Kinetics
Select the lecture you want to download:
1 Lecture 1: [Reactions Rate and Rate Laws]
Download PDF
2 Lecture 2: [Integrated Rate Laws and Half-Life Time]
Download PDF
Message


Execute


Cairo University - Chem
bot
4 Lecture 4: [Experimental Determination of Rate Law]
Download PDF
This message was sent automatically with n8n 13:06
Lecture Attendance Registration
Please enter the session passcode written on the board along with your Academic ID and Full Name in this exact format:
ATTEND <Passcode> <Academic_ID> <Full Name>
Example:
ATTEND KIN301 20231045 Ahmed Mohamed
This message was sent automatically with n8n 13:06
Lecture Attendance Registration
Please enter the session passcode written on the board along with your Academic ID and Full Name in this exact format:
Message


Cairo University - Chem
bot
5. Answer: (A) [A]0 / k
For a zero-order reaction:
[A]t = [A]0 - kt
Setting [A]t = 0 (complete consumption):
0 = [A]0 - kt → t = [A]0 / k
6. Answer: (C) ln(A)
Taking the natural logarithm of the Arrhenius equation:
ln(k) = ln(A) - (Ea/R) · (1/T)
This is a straight line (y = c + mx) where the intercept c is ln(A).
***
ملخص سريع بالعربي:
١. في تفاعلات الرتبة الثانية، الميل (slope) لرسمة 1/[A] مع الزمن بيساوي k.
٢. عمر النصف (t1/2) في الرتبة الأولى ثابت مبيعتمدش على التركيز الابتدائي.
٣. وحدة الـ k بتعرفنا الرتبة؛ لما تلاقي L²·mol⁻²·s⁻¹ يبقى التفاعل من الرتبة التالتة.
٤. زيادة الحرارة بتزود الـ k عشان بتزود عدد الجزيئات اللي عندها طاقة كافية تكسر حاجز طاقة التنشيط (Ea).
This message was sent automatically with n8n
Message

The Adopted Workflow ⟸ MOHAMED





Live Application

https://web.telegram.org/a/#8922398010

@chemoh_bot → Learn for all → Start you Experience with the chatbot





The Expected Outcome.

1- Transformation of Student Engagement.

- Replaces static e-learning constraints with dynamic conversational AI tutoring.
- Delivers a zero-friction learning interface accessible directly via the Telegram Bot API.

2- Curriculum & Knowledge Sync

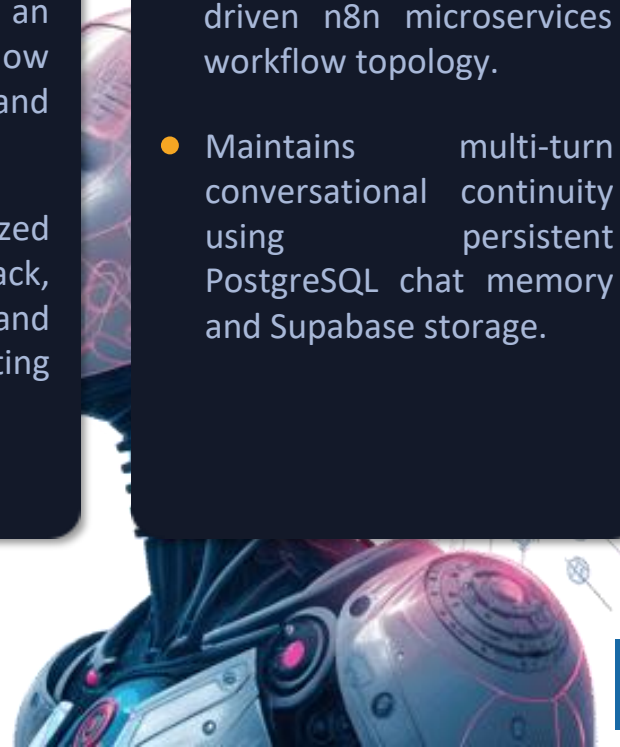
- Automates course synchronization through a background Google Drive ingestion pipeline.
- Converts and indexes curriculum text into 768-dimensional vector embeddings using Google Gemini.

3- Automated Adaptive Learning & Assessment.

- Eliminates manual grading bottlenecks by using an LLM evaluation workflow to assess student text and file submissions.
- Returning customized diagnostic feedback, awarding points, and dynamically updating student mastery tiers.

4- Scalable Architecture & Memory

- Operates on an event-driven n8n microservices workflow topology.
- Maintains multi-turn conversational continuity using persistent PostgreSQL chat memory and Supabase storage.



Project Roadmap/Roles

Phase 1

1 Week

Concept& Planning

Foundation Base:

- Core chatbot engine
- NLP integration
- Basic adaptive system
- Student`s Limitations

HOSSAM/OMNIA/MOHAMED

Phase 2

2:4 Weeks

Execution

System Building:

- N8n workflow
- LLM fine-tuning
- Multi-branches support
- Mobile chatbot

ALL TEAM MEMBERS.

Phase 3

1:2 Weeks

Implementation& Testing

Testing & Editing:

- N8n functionality
- Analytics API
- Chatbot functionality
- Bug Reports

MOHAMED

Phase 4

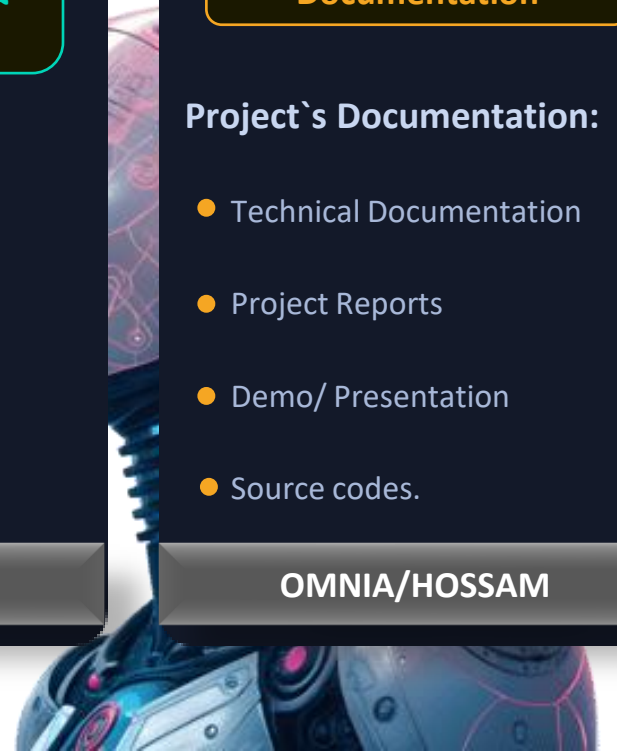
1:2 Weeks

Documentation

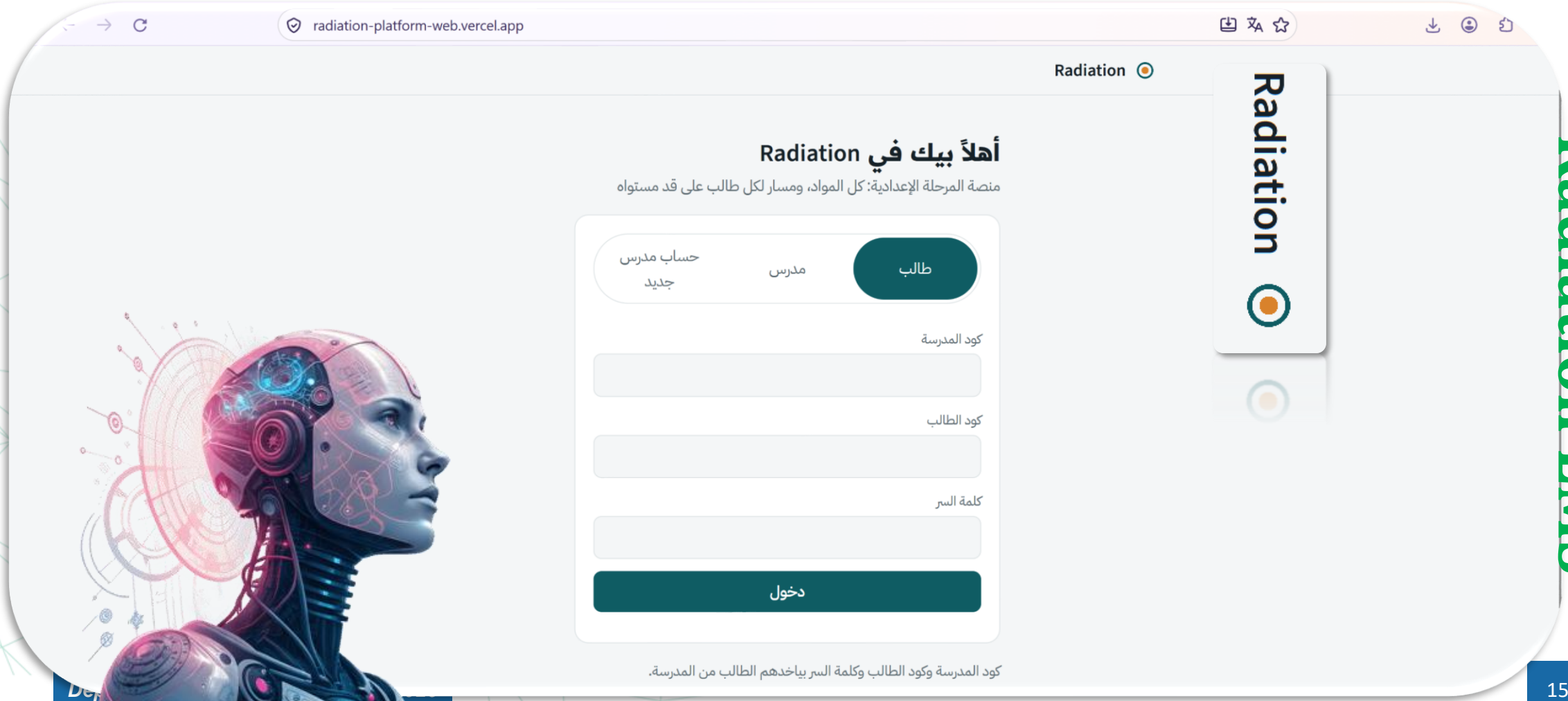
Project`s Documentation:

- Technical Documentation
- Project Reports
- Demo/ Presentation
- Source codes.

OMNIA/HOSSAM



Phase 5: Our Latest Work.



radiation-platform-web.vercel.app

Radiation

المجموعة | الواجبات | الجدول | التقارير | الحضور والفلوس | الفيديوهات | أسئلة الطلاب | إضافة طلاب

Omnia Reda Ghalwash خروج

تانية بكالوريا | الكيمياء

المجموعة

الكيمياء · مين ضعيف في إيه، على مستوى كل مفهوم في الدرس.

المجموعة: B · 2026/2027/2

الدرس: Changes of State 1.1



ملخص المجموعة

- 0 من 1 طالب ضعاف في مفهومين أو أكتر.
- 1 طالب لسه ما بدأوش الدرس ده.

الطالب	1	2	3	4	5
21021 shimaa mahmoud					
متوسط المجموعة					

1 Celsius and absolute temperature
2 Changes of state and the heating curve
3 Calculating the amount of heat
4 Intermolecular forces
5 Attractive forces and melting/boiling

رواد مصر الرقمية

Radiation LMS



form-web.vercel.app

Radiation

Hussien Mohammed خروج

النهارده | مساري | الدروس | الواجبات | المساعد | مساع

أهلاً يا Hussien

الخميس، ١٥ أكتوبر · تانية إعدادي

خطة النهارده · حوالي 13 دقيقة

1 Plasma: the fourth state of matter
العلوم · نقطة ضعف (0%) · 5 دقايق
تدرّب

2 Changes in the States of Matter
العلوم · درس جديد: اختبار تشخيصي قصير · 8 دقايق
ابدأ

كل يوم من واجباتك، ونقط ضعفك، والمراجعة ال



سياسة الخصوصية

form-web.vercel.app

Radiation

Hussien Mohammed خروج

النهارده | مساري | الدروس | الواجبات | المساعد | مساع

العلوم · اختار درس عشان تشوف مفاهيمه ومستواك في كل واحد.

لم يبدأ | ضعيف | بيتعلم | متقن

الوحدة 1 · Matter and Energy

1. States of Matter — 44 سؤال
0 من 8 مفهوم متقن

2. Changes in the States of Matter — 41 سؤال
0 من 8 مفهوم متقن

3. Internal Energy and Temperature — 36 سؤال
0 من 7 مفهوم متقن

4. Methods of Heat Transfer — 62 سؤال
0 من 11 مفهوم متقن

الوحدة 2 · Matter and Chemical Reactions

form-web.vercel.app

Radiation

Hussien Mohammed خروج

النهارده | مساري | الدروس | الواجبات | المساعد | مساع

تانية إعدادي | العلوم | الرياضيات | اللغة الإنجليزية

ري

تانية إعدادي

ة الجاية في مسارك

Matter and its physical states
متطلب لـ «Shape, fluidity and viscosity»
States of Matter · «
تدرّب

Particle theory of matter and modelling
متطلب لـ «Explaining properties using attraction and spacing»
States of Matter
تدرّب

Shape, fluidity and viscosity
متطلب لـ «Explaining properties using attraction and spacing»
States of Matter
تدرّب

Volume and compressibility
متطلب لـ «Explaining properties

Continuous Adaptation Loop

01

Input

Student asks a question or submits an answer via chat.

02

Understand

NLP parses intent, topic, and sentiment from the message.

03

Assess

Knowledge-tracing model updates the live student profile.

04

Adapt

Engine selects content and difficulty for the next step.

05

Respond

Assistant replies with explanation, hint, or new challenge.

Every response feeds the next — the loop repeats continuously, so the assistant gets smarter about the student with each interaction.

WHAT'S NEXT

Our Future Step (beyond the prototype)

NOW	NEXT	LATER	VISION
MVP **Minimum Viable Product** Working chatbot, core NLP, and adaptive Learning Materials on one subject.	**Scale** Developing Multi-subject content, Developing LMS, gaining richer knowledge tracing.	**Enrich** Voice interaction, multilingual (Arabic/English), mobile app.	**Platform** Open API for schools and universities to plug in their curricula.

Project Deliverables/Reports

- ✓ Project Description (Technical Documentation).
- ✓ Final Presentation (PPT/PDF).
- ✓ ZIP File for Source Code.
- ✓ Live Application Link.
- ✓ ***Project Documentation Reports:***
  - *1- Planning& Management Documentation.*
  - *2- Requirements Gathering.*
  - *3- System Analysis & Design.*
  - *4- Implementation &Testing Process.*
- ✓ Video Demo for 2 minutes.
- ✓ User Manual.
- ✓ The Project`s GitHub Link.



Let`s have learning that adapts

to every student.

🚀 Try the Demo

📊 View Reports

🌱 Join the Beta

- ***Contact information:***
- Hosam Hasan: *(Team Leader)* 01122333955 – hosamhasan88@cu.edu.eg
- Omnia Reda Ghalwash: 01221618561 – ghalwashomnia@gmail.com
- Mohammed Hussien: 01062374896 – Mohammedhussien23397@gmail.com

Thank you

DEPI Industry Graduation Project • Questions & Discussion Welcome